Review

Whitney Huang

Clemson University | DSA 8070

The Big Picture: Multivariate Analysis

Central idea: Analyze multiple variables jointly, accounting for their dependence and shared structure.

- **Explore:** What patterns and relationships are present?
- **Infer:** What can we conclude about populations?
- **Relate and predict:** How are variables or sets of variables connected?
- **Simplify and discover:** Can we reduce dimensions or identify groups?

The Common Statistical Foundation

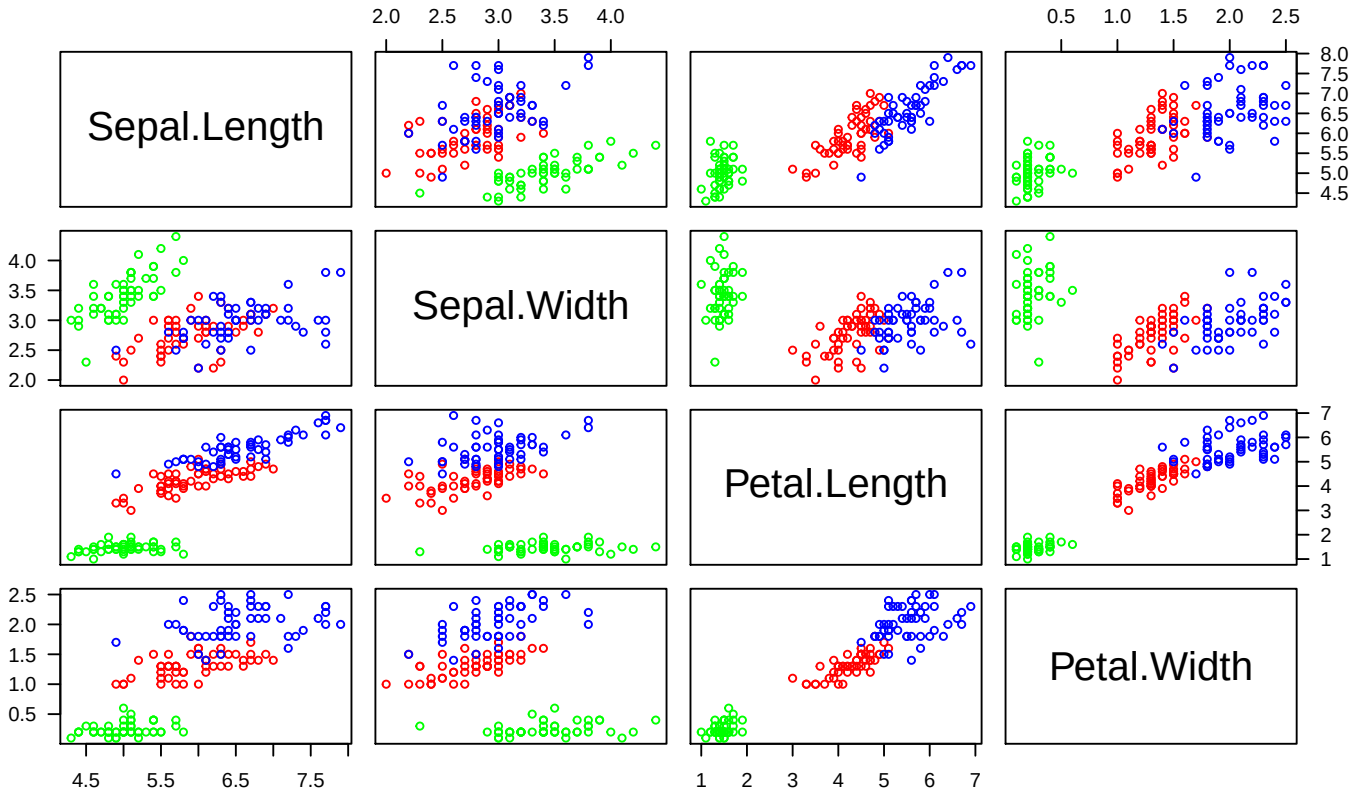
- **Data matrix:** $\boldsymbol{X} \in \mathbb{R}^{n \times p}$
- **Mean vector and covariance matrix:**

$$\bar{\mathbf{x}}, \qquad \mathbf{S} = \frac{1}{n-1} \sum_{i=1}^{n} (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^T$$

- **Geometry:** Eigenvalues, eigenvectors, quadratic forms, and Mahalanobis distance.
- **Probability models:** Multivariate normal distributions, dependence structures, and copula models.

Takeaway: Covariance connects multivariate geometry, inference, and modeling.

Start with Multivariate Data Exploration



Examine: Dependence • Group separation • Outliers

Explore before choosing a method.

Multivariate Inference: What Are We Testing?

- **Mean vectors:** Hotelling's T^2 and MANOVA.
- **Covariance structure:** Estimation, comparison, and high-dimensional challenges.
- **Joint inference:** Account for dependence among outcomes rather than testing each variable separately.

Practical question: Are the groups different in their *multivariate profiles*, rather than in just one variable?

Multivariate Regression: Relate and Predict

Question: How do predictors jointly explain multiple response variables?

$$\boldsymbol{Y} = \boldsymbol{X}\boldsymbol{B} + \boldsymbol{E}$$

- **Multivariate regression:** Model several responses simultaneously.
- **Reduced-rank regression:** Use a lower-rank coefficient matrix to simplify the predictor–response relationship.
- **Inference and prediction:** Evaluate effects, uncertainty, and predictive performance.

Key idea: Exploit relationships among multiple responses.

Canonical Correlation Analysis (CCA)

Question: How strongly are two sets of variables related?

$$U = \boldsymbol{a}^T \boldsymbol{X}, \qquad V = \boldsymbol{b}^T \boldsymbol{Y}$$

$$\max_{\boldsymbol{a},\boldsymbol{b}} \mathrm{Corr}(U, V)$$

- Find pairs of linear combinations with maximum correlation.
- Interpret canonical correlations, coefficients, and loadings.
- Unlike regression, CCA treats the two variable sets symmetrically.

Dimension Reduction: PCA vs. Factor Analysis

	PCA	**Factor Analysis**
Goal	Summarize variation	Explain covariance
Approach	Linear combinations of observed variables	Latent factors and variable-specific errors
Focus	Variance explained	Common variation and factor loadings
Interpret	Component scores and loadings	Factors, loadings, and communalities

Key distinction: PCA summarizes data; factor analysis models underlying latent structure.

Classification: Predicting Group Membership

Setting: Group labels are available for training observations.

- **Logistic regression:** Model class probabilities.
- **LDA and QDA:** Classify using distributional assumptions and covariance structure.
- **Support vector machines:** Construct separating decision boundaries.

Practical focus: Compare classification performance on held-out data, not only training accuracy.

Clustering: Discovering Groups

Setting: Group labels are unknown.

- **K-means:** Partition observations by minimizing within-cluster variation.
- **Hierarchical clustering:** Build nested groups using distances and linkage rules.
- **Gaussian mixture models:** Represent clusters probabilistically.

Key questions: How many clusters? How stable? Are the groups meaningful?

Classification vs. Clustering

	Classification	**Clustering**
Labels	Known in training	Unknown
Goal	Predict group membership	Discover group structure
Output	Predicted classes or probabilities	Cluster assignments or probabilities
Evaluation	Test-set accuracy, sensitivity, etc.	Separation, stability, interpretability

Remember: Finding clusters is not the same as predicting known classes.

Multidimensional Scaling: From Distances to Maps

Question: Can pairwise dissimilarities be represented in a low-dimensional space?

- **Classical MDS:** Use eigendecomposition to approximate Euclidean distance structure.
- **Non-metric MDS:** Preserve dissimilarity rankings by minimizing STRESS.
- **Interpretation:** Examine proximity, separation, and goodness of fit.

Key idea: The choice of dissimilarity determines what the map represents.

Beyond Linear Structure: Nonlinear Embedding

Motivation: Important data structure may lie on a curved, low-dimensional manifold.

- **Polynomial and kernel PCA:** Capture nonlinear variation through transformed feature spaces.
- **Isomap and LLE:** Use manifold geometry and local neighborhood relationships.
- **t-SNE and UMAP:** Emphasize neighborhood structure for visualization.

Caution: Different embeddings preserve different aspects of the data.

A Practical Multivariate Analysis Workflow

1. **Define the question:** Inference, prediction, association, grouping, or visualization?

2. **Explore and preprocess:** Missing values, scaling, outliers, and dependence.

3. **Choose and fit a method:** Match assumptions and objectives to the data.

4. **Evaluate and interpret:** Diagnostics, uncertainty, validation, and limitations.

Good analysis is more than running an R function.

Course Takeaways: Think Multivariately

- **Dependence matters.** Variables should not always be analyzed in isolation.
- **Methods serve different purposes.** Match the statistical question to the method.
- **Geometry connects many techniques.** Covariance, distances, projections, and eigenvectors recur throughout the course.
- **Interpretation matters most.** Understand what the method preserves, assumes, and potentially misses.

Final message: Understand the question, the data, and the assumptions before choosing the method.